

Cogging torque reduction by eccentric structure of teeth in external rotor permanent magnet synchronous motors

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Abstract: Aiming at the problem of torque ripple caused by cogging torque in the external rotor permanent magnet synchronous motor, this study proposes a method for reducing the cogging torque by the eccentric structure of stator teeth. Taking a 3 kW external rotor permanent magnet synchronous motor as an example, the validity of the method proposed is verified by theoretical analysis and finite element calculation. Finally, the other performances of the models with different teeth eccentricities are compared, and a prototype is manufactured to verify the correctness of the simulation results. The research shows that the eccentric structure of stator teeth does not affect the other performances of the motor greatly while reducing the cogging torque of external rotor permanent magnet synchronous motor effectively.

1 Introduction

The external rotor permanent magnet synchronous motor has been widely used in electric vehicles, flywheel energy storage and other fields due to its large moment of inertia, good thermal performance and high power density [1–3]. However, as a permanent magnet motor, the development of external rotor permanent magnet synchronous motor has been restricted by the problem of cogging torque. Excessive cogging torque causes acoustic noise and speed ripples, which will affect the operating performance of the motor seriously. Therefore, it is extremely important to reduce cogging torque as much as possible without affecting other operating performances of the motor greatly.

Research on reducing the cogging torque of permanent magnet motor has always been paid attention by scholars and motor designers. The authors of [4–8] deduce the expression of cogging torque and analyse the impact of pole arc coefficient, slot number and pole number combination, the width of stator slot, the eccentricity of magnet pole as well as stator auxiliary slot on cogging torque based on the expression. Wegiel [9] propose a simplified energy method based on the motor with a permanent magnet located on the surface of the rotor, and analyse the influence of the pole composed of the segmented permanent magnet on the cogging torque. Yang and Wang [10] investigate the effect of permanent magnets asymmetry on the cogging torque of interior permanent magnet motor and proposes a method for calculating the shifting angles of permanent magnets that can effectively reduce the cogging torque. In [11], two models with uniformed teeth and non-uniformed teeth are proposed to study the influences of stator tooth width and slot-pole match on the cogging torque. The magnet skewing structure can reduce cogging torque to a satisfactory level [12–14], but it is difficult to assemble and has a narrow range of applications. In order to reduce the cogging torque, the shape of the rotor poles and stator teeth are optimised by the topological gradient method in [15]. Wang *et al.* [16]

propose a new technique about rotor tooth axial pairing to reduce the cogging torque of permanent magnet flux-switching machines. The validity of the technique is confirmed by the method of finite element analysis and experiment. For the line-start permanent synchronous motor, Tang *et al.* [17] propose an equivalent method of the rotor magneto motive force produced by permanent magnets, and analyses the effect of the combination of stator slot number and pole number as well as stator slot skewing on cogging torque. Li *et al.* [18], Liu *et al.* [19] reduce the cogging torque by shifting the notch position. In [20], the Taguchi robust design is proposed to reduce the cogging torque of the permanent magnet synchronous motor in the application of electric power steering systems. However, the existing methods of cogging torque reduction are mainly for internal rotor motors. For external rotor motors, the above methods may not be applicable.

This paper derives the expression of cogging torque for external rotor permanent magnet synchronous motor by energy method and Fourier decomposition [6]. The influence of eccentric structure on the cogging torque is investigated according to the analytical expression and structure of the motor. The finite-element analysis is adopted to verify the correctness of the analytical method.

2 Cogging torque analysis by energy method

2.1 Motor structure

As shown in Fig. 1, the internal stator is stationary, and the rotor rotates with the shaft. The permanent magnetic poles are located between the rotor core and the rotor cover. In the assembly process, the rotor cover protects the permanent magnetic poles from damage. Compared with the same size of the inner rotor motor, the structure features of the external rotor motor make it possible to have a larger air gap, better thermal performance and higher moment of inertia [2].

2.2 Cogging torque expression

Cogging torque is the torque generated due to the interaction between the permanent magnets and the slotting on the stator. In order to derive the expression of the cogging torque conveniently, the following assumptions are made in theoretical analysis: (i) the influence of magnetic circuit saturation is neglected; (ii) the permeability of core tends to be infinite; (iii) the permeability of permanent magnets is the same as that of the air; (iv) the energy in the permanent magnet is fixed [7].

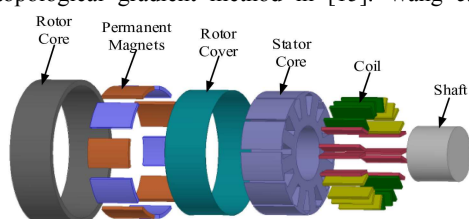


Fig. 1 Motor construction

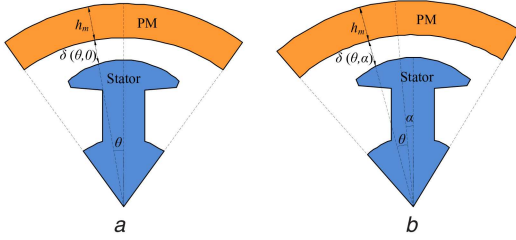


Fig. 2 Position diagram of stator and rotor

(a) When the centreline of the stator coincides with the centreline of the rotor ($\alpha = 0$),
(b) When the centreline of the stator does not coincide with the centreline of the rotor ($\alpha \neq 0$)

When the rotor moves relative to the stator, the reluctance between the permanent magnets and the stator teeth corresponding to the permanent magnets remains constant. Since the energy in the permanent magnet is assumed to be a fixed value, the magnetic field energy of the pole arc part of permanent magnets remains constant. However, the reluctance of the stator core area corresponding to both sides of the permanent magnet is mutative, which will lead to the changes in the energy in the magnetic field, and cause cogging torque. Therefore, the cogging torque can be expressed as the negative derivative of the magnetic co-energy in the motor [5]

$$T_{\text{cog}} = -\frac{\partial W(\alpha)}{\partial \alpha} \quad (1)$$

where $W(\alpha)$ is the magnetic energy, and α is the angle between a specified permanent magnet centreline and a specified stator tooth centreline.

Assuming that the permeability of the core is infinite, the stored energy of the motor magnetic field can be approximated as the magnetic energy in the permanent magnets and the air gap. At the same time, since the energy in the permanent magnets is assumed to be a constant, the magnetic field energy which generates cogging torque can be expressed as

$$W(\alpha) \simeq W_{\text{gap}}(\alpha) = \frac{1}{\mu_0} \int_V |\mathbf{B}(\theta, \alpha)|^2 dV \quad (2)$$

where μ_0 is the permeability of the air; $\mathbf{B}(\theta, \alpha)$ is the air gap flux density in the direction.

The distribution expression of air gap flux density in the circumferential direction is

$$\mathbf{B}(\theta, \alpha) = \mathbf{B}_r(\theta) \frac{h_m}{h_m + \delta(\theta, \alpha)} \quad (3)$$

where $\mathbf{B}_r(\theta)$ is the distribution of residual flux density in the air gap, assuming the position $\theta = 0$ is the centre of any motor pole position; h_m is the length in permanent magnet magnetisation direction; $\delta(\theta, \alpha)$ is the air gap length distributed along the circumference of the motor when the angle between magnetic pole centreline and tooth centreline is α .

Substitute (3) into (2)

$$W(\alpha) = \frac{1}{2\mu_0} \int_V |\mathbf{B}_r(\theta)|^2 \left(\frac{h_m}{h_m + \delta(\theta, \alpha)} \right)^2 dV \quad (4)$$

We can transform (4) and get the expression of cogging torque through the Fourier decomposition of $|\mathbf{B}_r(\theta)|^2$ and $(h_m/(h_m + \delta(\theta, \alpha)))^2$. The cycle of $|\mathbf{B}_r(\theta)|^2$ is π/p , and the cycle of $(h_m/(h_m + \delta(\theta, \alpha)))^2$ is $2\pi/z$ [11]. Where p means the number of pole pairs, z is the number of stator slots. The Fourier expansions of $|\mathbf{B}_r(\theta)|^2$ is shown as

$$|\mathbf{B}_r(\theta)|^2 = B_{r0} + \sum_{n=1}^{\infty} B_{rn} \cos(2np\theta) \quad (5)$$

where

$$B_{r0} = \frac{p}{\pi} \int_{-\pi\alpha_p/2p}^{\pi\alpha_p/2p} |\mathbf{B}_r(\theta)|^2 d\theta \quad (6)$$

$$B_{rn} = \frac{2p}{\pi} \int_{-\pi\alpha_p/2p}^{\pi\alpha_p/2p} |\mathbf{B}_r(\theta)|^2 \cos(2np\theta) d\theta \quad (7)$$

where α_p is the pole arc coefficient.

Fig. 2 is the position diagram of stator and rotor. When $\alpha = 0$, the Fourier expansions of $(h_m/(h_m + \delta(\theta, 0)))^2$ can be expressed as

$$\left(\frac{h_m}{h_m + \delta(\theta, 0)} \right)^2 = C_0 + \sum_{n=1}^{\infty} C_n \cos(nz\theta) \quad (8)$$

Integrate (8) on θ from $-\pi/z$ to π/z .

$$\begin{aligned} \int_{-\pi/z}^{\pi/z} \left(\frac{h_m}{h_m + \delta(\theta, 0)} \right)^2 d\theta &= \int_{-\pi/z}^{\pi/z} C_0 d\theta + \sum_{n=1}^{\infty} C_n \int_{-\pi/z}^{\pi/z} \cos(nz\theta) d\theta \\ &= \frac{2\pi}{z} C_0 \end{aligned} \quad (9)$$

$$C_0 = \frac{z}{2\pi} \int_{-\pi/z}^{\pi/z} \left(\frac{h_m}{h_m + \delta(\theta, 0)} \right)^2 d\theta \quad (10)$$

Integrate (8) on θ from $-\pi/z$ to π/z after it is multiplied by $\cos(kz\theta)$.

$$\begin{aligned} \int_{-\pi/z}^{\pi/z} \left(\frac{h_m}{h_m + \delta(\theta, 0)} \right)^2 \cos(kz\theta) d\theta &= C_0 \int_{-\pi/z}^{\pi/z} \cos(kz\theta) d\theta + \sum_{n=1}^{\infty} C_n \int_{-\pi/z}^{\pi/z} \cos(nz\theta) \cos(kz\theta) d\theta \\ &= \sum_{n=1}^{\infty} C_n \int_{-\pi/z}^{\pi/z} \frac{1}{2} [\cos((nz + kz)\theta) + \cos((nz - kz)\theta)] d\theta \end{aligned} \quad (11)$$

It is easily to find that

$$\int_{-\pi/z}^{\pi/z} \cos((nz + kz)\theta) d\theta = 0$$

When $n \neq k$

$$\int_{-\pi/z}^{\pi/z} \cos((nz - kz)\theta) d\theta = 0$$

When $n = k$

$$\int_{-\pi/z}^{\pi/z} \cos((nz - kz)\theta) d\theta = \frac{2\pi}{z}$$

$$C_n = \frac{z}{\pi} \int_{-\pi/z}^{\pi/z} \left(\frac{h_m}{h_m + \delta(\theta, 0)} \right)^2 \cos(nz\theta) d\theta \quad (12)$$

When $\alpha \neq 0$, the expression $(h_m/(h_m + \delta(\theta, \alpha)))^2$ can be obtained by rotating angle α to (8)

$$\left(\frac{h_m}{h_m + \delta(\theta, \alpha)} \right)^2 = C_0 + \sum_{n=1}^{\infty} C_n \cos(nz(\theta + \alpha)) \quad (13)$$

The C_0 and C_n in (13) are equal to the C_0 and C_n in (8).

Equation (14) can be obtained by combining (1) and (4)–(13)

$$T_{\text{cog}}(\alpha) = \frac{\pi z L_{Fe}}{4\mu_0} (R_1^2 - R_2^2) \sum_{n=1}^{\infty} n C_n B_r(nz/2p) \sin(nz\alpha) \quad (14)$$

where L_{Fe} is the effective axial length of the motor; R_1 is the inner radius of the rotor; R_2 is the outer radius of the stator; n is a positive integer that guarantees $nz/2p$ to be an integer.

3 Effects of eccentric structure on cogging torque

The expression of cogging torque shows that after $|B_r(\theta)|^2$ is decomposed, only the $nz/2p$ harmonic will have an effect on the cogging torque. Cutting down the $nz/2p$ harmonic can reduce the cogging torque of the external rotor permanent magnet synchronous motor effectively [10]. Therefore, according to the structure of the external permanent magnet synchronous motor, a method of reducing cogging torque by cutting down the $nz/2p$ harmonic with the stator teeth eccentric structure is proposed.

3.1 Eccentric structure of stator teeth

The eccentric structure of stator teeth is shown in Fig. 3. The centre of tooth arc is at the point O when the stator teeth do not adopt eccentric structure. Under this circumstance, the length of the air gap corresponding to the stator teeth is $\delta(\theta, \alpha)$, which is a constant. When the eccentric structure is used in stator teeth, tooth arc centre moves from the point O to the point O' , the length of the air gap under the stator teeth is $\delta'(\theta, \alpha)$, which will change as θ changes. h is the eccentricity of the stator teeth.

3.2 Effect of eccentric structure

When the eccentric structure is not adopted in stator teeth, the air gap flux density expression is

$$B(\theta, \alpha) = B_r(\theta) \frac{h_m}{h_m + \delta(\theta, \alpha)} \quad (15)$$

When the stator teeth adopt the eccentric structure, the expression of the air gap flux density is

$$\begin{aligned} B'(\theta, \alpha) &= B_r(\theta) \frac{h_m}{h_m + \delta'(\theta, \alpha)} \\ &= \frac{h_m + \delta(\theta, \alpha)}{h_m + \delta'(\theta, \alpha)} B_r(\theta) \frac{h_m}{h_m + \delta(\theta, \alpha)} \\ &= B_r'(\theta, \alpha) \frac{h_m}{h_m + \delta(\theta, \alpha)} \end{aligned} \quad (16)$$

In (16), the $B_r'(\theta, \alpha)$ is expressed as

$$B_r'(\theta, \alpha) = B_r(\theta) \frac{h_m + \delta(\theta, \alpha)}{h_m + \delta'(\theta, \alpha)} \quad (17)$$

The residual flux density in the motor with eccentric structure can be expressed by the residual flux density in the motor with non-eccentric structure. Compared with the non-eccentric structure of the teeth, the teeth with eccentric structure will only affect the $B_r'(\theta, \alpha)$ in (16), which is significance to compare with the $nz/2p$ harmonic after the $|B_r'(\theta, \alpha)|^2$ is decomposed.

According to Fig. 3, the expression of the air gap length corresponding to the stator teeth can be obtained

$$\delta'(\theta, \alpha) = \delta(\theta, \alpha) + R_2 - h \cos \theta - \sqrt{(R_2 - h)^2 - (h \sin \theta)^2} \quad (18)$$

Combining (17) and (18)

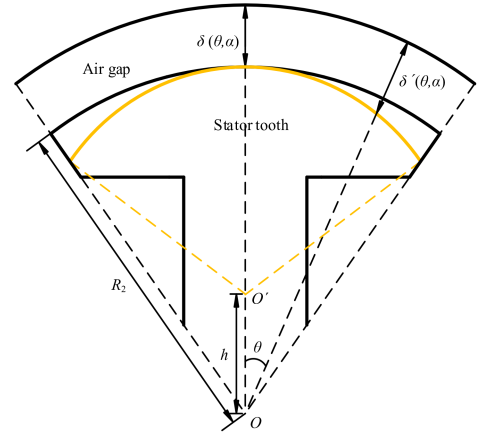


Fig. 3 Schematic diagram of the eccentric structure

Table 1 Parameters of external rotor permanent magnet synchronous motor

Parameter	Value	Parameter	Value
stator outer diameter, mm	134	stator inner diameter, mm	60
rotor outer diameter, mm	156	rotor inner diameter, mm	136
magnet thickness, mm	4	residual flux density, T	1.23
pole pairs	5	air gap length, mm	1
axial length, mm	43	number of stator slots	12
pole arc coefficient	0.716	coercive force, kA/m	860

$$\begin{aligned} B_r'(\theta, \alpha) &= \frac{(h_m + \delta(\theta, \alpha)) B_r(\theta)}{h_m + \delta(\theta, \alpha) + R_2 - h \cos \theta - \sqrt{(R_2 - h)^2 - (h \sin \theta)^2}} \end{aligned} \quad (19)$$

The expression of $|B_r'(\theta, \alpha)|^2$ is

$$|B_r'(\theta, \alpha)|^2 = \left(\frac{t(\theta, \alpha) B_r(\theta)}{t(\theta, \alpha) + 1 - b \cos \theta - \sqrt{(1 - b)^2 - (b \sin \theta)^2}} \right)^2 \quad (20)$$

where $t(\theta, \alpha) = (h_m + \delta(\theta, \alpha))/R_2$, $b = h/R_2$

This paper only needs to demonstrate the effect of eccentric structure on reducing the $nz/2p$ harmonic. Meanwhile, the eccentric structure does not affect $t(\theta, \alpha)$ and $B_r(\theta)$ in (20). Since the value of α only affects the range of the value of θ in (20). For the convenience of calculation, $\alpha = 0$ is taken. The effectiveness of the method can be proved by comparing the content of $nz/2p$ harmonic in $|B_r'(\theta, 0)|^2$ with different eccentricities according to the parameters of the prototype.

The parameters of the prototype adopted in this paper are shown in Table 1. In order to find out the relationship between eccentricity and cogging torque more accurately, a large number of motor models with different eccentricities have been established for analysis. Due to the space limitations, we only select four kinds of motor models with an eccentricity of 0, 24, 28 and 32 mm, which can reflect the change rule of cogging torque on eccentricity.

Substituting the parameters into (20), we can get the distribution curves of $|B_r'(\theta, 0)|^2$ which are shown in Fig. 4.

Table 1 shows that $nz/2p = 6n/5$. To make sure that $nz/2p$ is an integer; n take a multiple of 5, then $nz/2p = 6k$. k is a positive integer. Through the Fourier decomposition of $|B_r'(\theta, 0)|^2$ in different eccentricities, we can get the comparison map of $6k$ harmonics which is shown in Fig. 5. The content of $6k$ harmonic decreased with the increasing of the eccentricity. Combining (14), we can prove that the eccentric structure of the stator teeth can effectively reduce the cogging torque of the external rotor permanent magnet synchronous motor. In the four kinds of motor

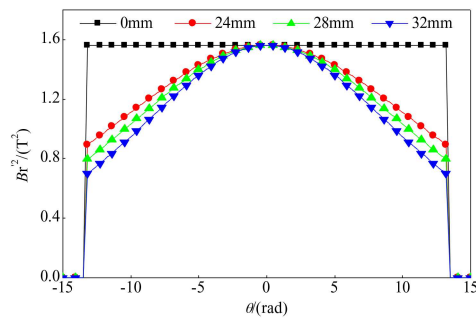


Fig. 4 Distribution curve of the square of residual flux density with different eccentricities

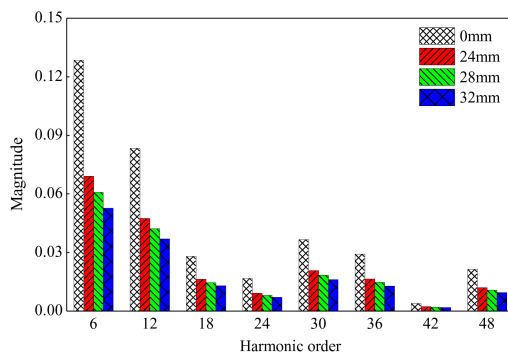


Fig. 5 Contrast of 6k harmonic with all the four cases

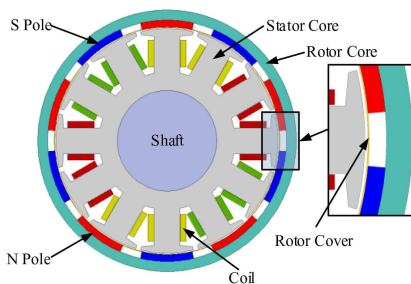


Fig. 6 Physical model of solved regional

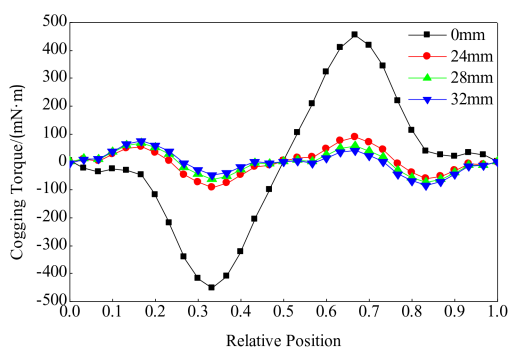


Fig. 7 Cogging torque for different eccentricities

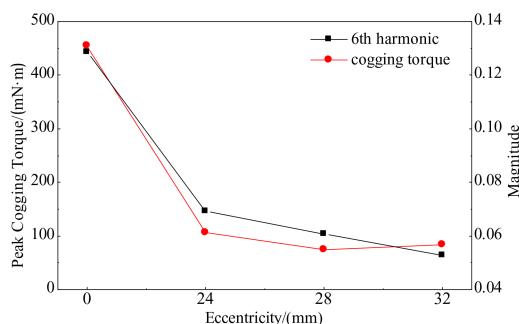


Fig. 8 Variation of cogging torque and sixth harmonics with eccentricity

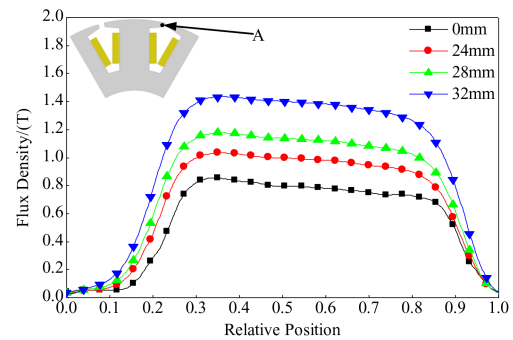


Fig. 9 Magnetic density on the edge of the tooth for different eccentricities

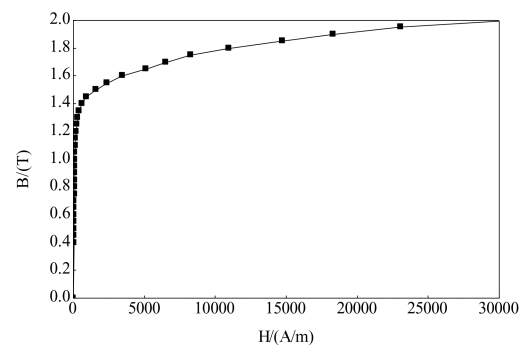


Fig. 10 Magnetisation curve of stator core material

models analysed, the cogging torque of the model with an eccentricity of 32 mm is the minimum.

4 Finite element analysis

To demonstrate the correctness of the method, the cogging torque of the above models is analysed by the finite element method. Fig. 6 is a two-dimensional physical model of the motor when the stator teeth do not adopt the eccentric structure.

Fig. 7 shows the cogging torque of the finite element analysis. Compared with the motor model without eccentric structure, the cogging torque of the motor models with an eccentricity of 24, 28 and 32 mm is reduced by 80.3, 84.3 and 82.4%, respectively. The cogging torque reaches to the minimum when the eccentricity is 28 mm.

Combined with the motor parameters and (14), we can find that the change rule of 6k harmonics on eccentricity can reflect the change rule of cogging torque on eccentricity in theoretical analysis and Fig. 5 shows that 6k harmonics have the same change rule on eccentricity. To simplify the analysis process, we only observe the change rule of the sixth harmonic on eccentricity. The variation of cogging torque and sixth harmonic with eccentricity is shown in Fig. 8, which can compare the change rule of cogging torque on eccentricity between theoretical analysis and simulation results.

From Fig. 8, we find that the cogging torque in the finite element simulation decreases first, then increases with the increase of eccentricity, and reaches the minimum when the eccentricity reaches 28 mm; the sixth harmonic in the theoretical analysis will gradually decrease with the increase of the eccentricity, which means that the cogging torque in the theoretical analysis will gradually decrease with the increase of the eccentricity.

There is a slight deviation between finite element simulation and the theoretical analysis. From Fig. 3, we can see that the thickness of the teeth edge will gradually reduce with the eccentricity increases, so the excessive eccentricity may cause magnetic saturation at the edge of the teeth, and affect the simulation results.

To verify the correctness of the above assumption, this paper takes a point A at the edge of the stator tooth and calculates the curve of magnetic flux density with position by the method of finite element analysis. The simulation result is shown in Fig. 9, and Fig. 10 is the magnetisation curve of the stator core material.

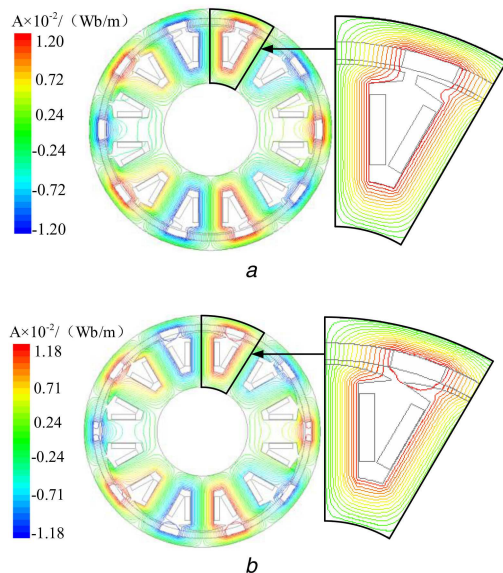


Fig. 11 Magnetic flux line distribution

(a) Magnetic flux line distribution when the teeth are not eccentric, (b) Magnetic flux line distribution when the eccentricity is 32 mm

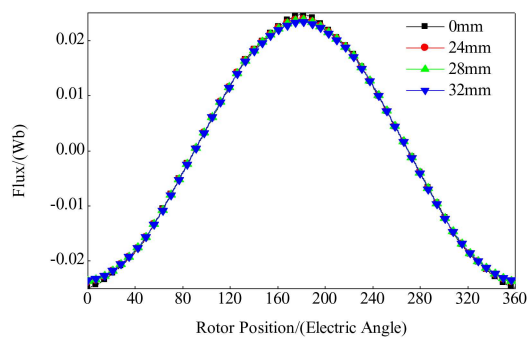


Fig. 12 Tooth flux for different eccentricities

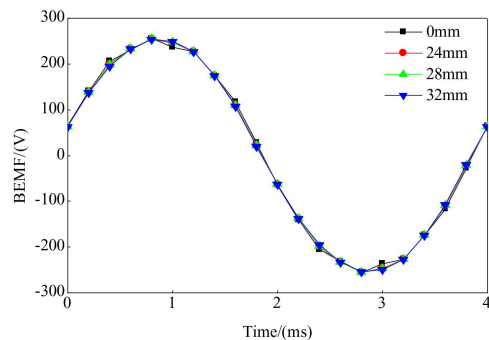


Fig. 13 BEMF waveforms for different eccentricities

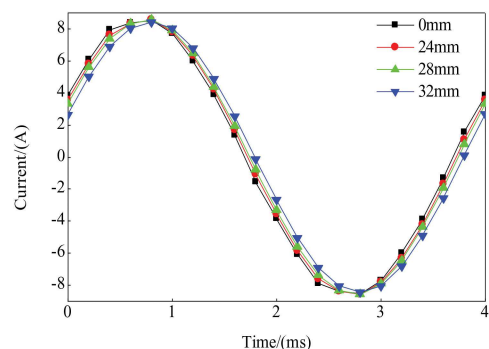


Fig. 14 Armature current waveforms for different eccentricities

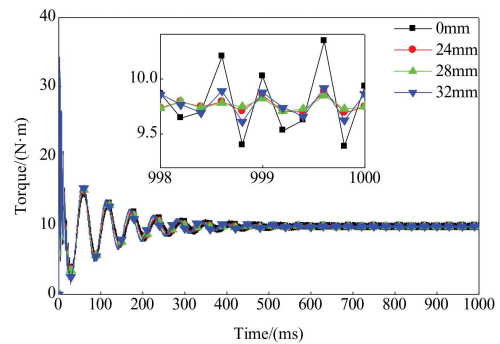


Fig. 15 Electromagnetic torque for different eccentricities

Figs. 9 and 10 show that the maximum flux density at the edge of the stator tooth reaches 1.436 T in the motor model with 32 mm in eccentricity, which reaches the saturation area of the magnetisation curve. When the magnetic field of the stator teeth is saturated, the permeability of the stator teeth will increase, and magnetic energy of stator teeth cannot be ignored as theoretical analysis, which leads to the discrepancy between simulation results and theoretical analysis.

To sum up, the appropriate stator tooth eccentricity can reduce the cogging torque of the external rotor permanent magnet synchronous motor effectively.

5 Effects of eccentric structure on performance

The eccentric structure of the stator teeth changes the structure of the motor, which will have an impact on the electromagnetic properties of the motor. Therefore, this section mainly studies the influence of the eccentric structure on the performance of the motor.

The eccentric structure will change the distribution of the air gap and affect the distribution of magnetic flux lines in the air gap. The distribution of magnetic flux lines is shown in Fig. 11, in which good correlations are observed. However, there is a small amount of magnetic flux lines passing the adjacent stator teeth through the air gap at the edge of stator teeth.

Changes in the distribution of magnetic lines will change the stator teeth flux, thereby affecting the back electromotive force (BEMF) and armature current of the motor. Fig. 12 is the flux of the single tooth in one cycle. Compared with the motor model with non-eccentric structure, the maximum magnetic flux per single tooth of the motor model with an eccentricity of 24, 28 and 32 mm decreased by 3.23, 4.05 and 5.26%, respectively, which proves that the eccentric structure has little effect on the magnetic flux of the stator teeth.

The waveforms of BEMF and armature current of phase A in one cycle under rated operation are shown in Figs. 13 and 14. It is shown that the BEMF and armature current of motor models with different eccentricities have a little difference in amplitude, phase and waveform, which indicates that the influence of eccentric structure on BEMF and armature current is also very weak.

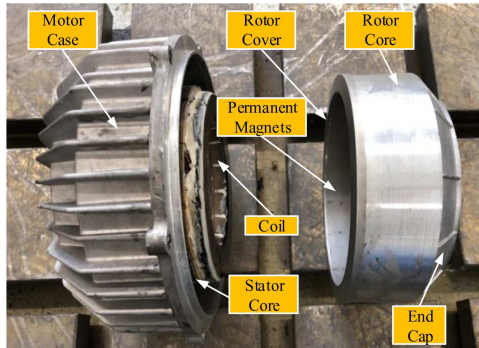
In addition, the increase of reluctance in the air gap will reduce the air gap flux density of motor, which would reduce the outputting power of the motor. Fig. 15 shows the electromagnetic torque curves of the motor with different eccentricities, in which good consistency is observed. The average values of the torque are 9.781, 9.765, 9.763 and 9.770 N·m. Therefore, the method of reducing cogging torque proposed in this paper will not have a greater effect on the output power of the motor.

Since the cogging torque as the main reason for the fluctuation of electromagnetic torque is reduced effectively, the electromagnetic torque ripple will also be reduced. From Fig. 15, it is shown that the electromagnetic torque ripple of the model with eccentric structure reduces by 78.3, 83.9, 69.1% of the original.

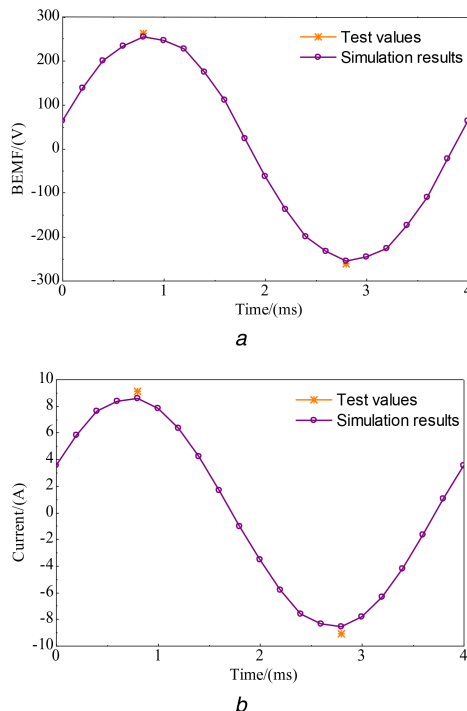
Table 2 compares loss and efficiency of the models with different eccentricities. The eccentric structure does not have a large impact on core loss, but the eddy current loss increases significantly as the eccentricity increases. Since the eddy current loss which mainly exists in permanent magnet and rotor cover is

Table 2 Comparison of loss and efficiency

Eccentricity, mm	Eddy current loss, W	Core loss, W	Efficiency, %
0	35.68	60.56	88.22
24	80.32	57.74	87.42
28	104.92	56.79	86.87
32	145.14	55.58	86.02

**Fig. 16** Structure of the prototype**Table 3** Comparison between the test results of the prototype and the simulation results of the motor with the eccentricity of 24 mm

	BEMF amplitude, V	Current amplitude, A	Efficiency, %
simulation value	262.75	8.61	87.42
measured value	261.67	9.09	87.03

**Fig. 17** Comparison of BEMF and armature current between test and simulation

(a) BEMF waveforms, (b) Armature current waveforms

relatively small, it would not have a greater impact on efficiency. In addition, the effect on the efficiency can be further reduced by dividing the permanent magnet and using the material of rotor cover and a permanent magnet with lower magnetic permeability.

In general, the effect of the eccentric structure on other performance of the motor is very small compared to the weakening effect of the eccentric structure on cogging torque and torque ripple.

6 Experimental results

The theoretical analysis results are obtained under the precondition of magnetic unsaturation, but the results of the finite element analysis show that the excessive eccentricity will cause magnetic saturation at the edge of the teeth, and lead to a deviation from the theoretical analysis. In order to ensure the performance of the prototype is not affected by the magnetic saturation, and to verify the correctness of the simulation results, a prototype with the eccentricity of 24 mm is manufactured, and the rated-load experiment is tested. The structure of the prototype is shown in Fig. 16. Table 3 is the comparison between the test results of the prototype and the simulation results of the motor model with the eccentricity of 24 mm.

The comparison of BEMF and armature current of phase A between test and simulation is shown in Fig. 17. Furthermore, the amplitudes of BEMF and armature current are listed in Table 3. Table 3 also lists the efficiency of the motor in test and simulation.

Fig. 17 and Table 3 show that the simulation results are in great agreement with the test results, but there is still a tiny discrepancy in the amplitudes of current. Considering the calculation deviation of software and the mechanical tolerances in manufacturing and assembly process, we can prove the correctness of the simulation results, which means that the eccentric structure can effectively reduce the cogging torque in external rotor permanent magnet synchronous motor.

7 Conclusions

This paper adopts the theoretical analysis and finite element analysis method to investigate the influence of the stator tooth eccentric structure on the cogging torque of the external rotor permanent magnet synchronous motor. The following conclusions can be obtained:

- The eccentric structure on stator teeth can effectively reduce the cogging torque, suppress the fluctuation of electromagnetic torque, and improve the stability of the external rotor permanent magnet synchronous motor.
- Excessive eccentricity will lead to the saturation of the magnetic circuit at the edge of teeth, which will increase the cogging torque slightly.
- The eccentric structure of stator teeth has little influence on the other operating performances of the motor and can reduce the fluctuation of the electromagnetic torque.
- When selecting the eccentricity, the influence of eccentric structure on the motor efficiency should be considered.

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